

COMP3242 Week 2 Practice Exam (Solutions)

Q1

- (a) $z = a^T x + b = 0.5 \cdot 1 + (-1) \cdot 2 + 0.2 = -1.3$.
- (b) $\sigma(-1.3) = \frac{1}{1+e^{1.3}} \approx 0.214$.
- (c) $L = -\log(0.214) \approx 1.54$.

Q2

A single linear classifier defines one hyperplane, so it can only separate data that are linearly separable. XOR is not linearly separable in input space. A two-layer perceptron can create intermediate nonlinear features in the hidden layer, then separate those transformed features in the output layer.

Q3

$$\#A = 8 \times 5 = 40, \#b = 8, \#C = 3 \times 8 = 24, \#d = 3.$$

Total parameters:

$$40 + 8 + 24 + 3 = 75.$$

Q4

- (a) Softmax + CE: single-label multi-class classification (e.g., classifying an image into exactly one of 10 classes).
- (b) Sigmoid + BCE: multi-label classification (e.g., one image may contain multiple attributes simultaneously).

Difference: softmax models mutually exclusive classes (probabilities sum to 1), while sigmoid treats each class as an independent binary decision.

Q5

$$\begin{aligned} z &= xA^T + b : (64, 20)(20, 50) + (50,) \Rightarrow (64, 50). \\ \text{logits} &= zC^T + d : (64, 50)(50, 10) + (10,) \Rightarrow (64, 10). \end{aligned}$$

For 'CrossEntropyLoss', label shape should be '(64,)' with integer class indices in '[0,9]' and dtype 'torch.long'.